

Employment

Software Engineer, Compiler

May 2025 – Present

RelationalAI, Logic Engine team

Remote

- Design and implement the procedural extension to RelationalAI’s query language
 - Scale adoption from internal algorithms to several enterprise workloads (15k queries/day)
 - Unlock previously inexpressible customer algorithms, which now run in under an hour
- Lead migration of legacy customer code (>300k LoC) to our Python library
 - Build a compiler toolchain to incrementally transpile legacy code
 - Accelerate benchmark workloads up to 30% by eliminating costly legacy compiler passes
- Deliver GQL-style graph pattern matching as sole compiler engineer with the Symbolic AI team
- Maintain the database engine’s Protocol Buffer-based entry point (100k+ queries/day)

Graduate Researcher

Sep 2022 – Apr 2025

Northeastern University, Programming Research Lab

Boston, USA

- Designed a language to express optimization problems under uncertainty [1]
 - Implemented a novel binary decision diagram algorithm to compute optimal values
 - Outperformed a state-of-the-art probabilistic programming language by up to 10x
- Co-led the first empirical study of online proof assistant communities [2]

R&D Intern

May 2024 – Dec 2024

RelationalAI, Logic Engine team

Remote

- Shipped a decompiler that lifts optimized Datalog back to the source query
- Integrated it into the production query profiler to automatically flag performance bottlenecks

Education

Master of Science, Computer Science

Sep 2022 – May 2025

Northeastern University

Boston, USA

Bachelor of Science, Mathematics and Philosophy

Aug 2018 – May 2022

Carnegie Mellon University

Pittsburgh, USA

Skills

Programming languages: Julia, Python, Datalog, OCaml, Rust, Go, Prolog, C/C++, Haskell

Tools: Unix, Git, Docker, Datadog, Observe, Snowflake, DuckDB, Souffle, LLVM/MLIR

Awards and Recognition

First Place, ACM Student Research Competition at PLDI (top PL conference)

2023

Scholarship Recipient, Programming Languages Mentoring Workshop (PLMW) at PLDI

2023

University and College Honors, Carnegie Mellon University

2022

Publications

- [1] Minsung Cho, John Gouwar, and Steven Holtzen. “Scaling Optimization over Uncertainty via Compilation”. In: *Proc. ACM Program. Lang. (OOPSLA)* (2025). DOI: 10.1145/3720500.
- [2] Gwenyth Lincroft*, Minsung Cho*, et al. “Thirty-Three Years of Mathematicians and Software Engineers: A Case Study of Domain Expertise and Participation in Proof Assistant Ecosystems”. In: *21st IEEE/ACM International Conference on Mining Software Repositories, MSR 2024*. ACM, 2024, pp. 1–13. DOI: 10.1145/3643991.3644908. *Joint first authors.